17. Stability in LTI systems

If we have a mathematical model of our system, then we can test for stability by exciting all frequencies at the input using a delta function. We generally find two cases:

Stable. When the impulse is applied to the system, the output is characterised by transient behaviour (sometimes with oscillation) which reduces over time leaving the system back in a quiescent state.

Unstable. The transients do not reduce but actually increase with time, such that the quiescent condition is never returned to.

There is a third case on the limit between these two: the output oscillates infinitely (similar to the un-damped harmonic oscillator in Section 14), or moves to a new constant output. As long as the output does not grow, this is still considered stable.

17.1. Stability and feedback

We briefly looked at stability in the context of feedback amplifiers. We saw how a negative feedback system can go into unstable positive feedback when the phase change at the output tends to 180°. In an amplifier this is very bad, since such feedback then tends to increase the gain rather than decrease it. The output tends to grow until it reaches some physical limit, usually V_{ss} (the supply voltage). We will now investigate this in more detail. Consider again the equation for an amplifier wired for negative feedback:

$$G = \frac{A}{1+\beta A}\,,$$

where A and G are variables which we saw were a function of frequency and β is the feedback fraction. We stated that we must have $A \ll 1$ as the output phase $\phi \to -180°$. For the purposes of stability analysis we assume the voltage follower configuration which means that *all* the output is fed back to the input (this is the worst-case situation for stability analyses) so $\beta = 1$. For the case $\phi = -180°$ negative feedback becomes positive feedback and we can rewrite G as

$$G = \frac{A}{1-A}\,.$$

We must avoid $A = 1$ and $\phi = -180°$ since then $G \to \infty$. We will now do this more formally for the general case of a transfer function $G(s)$.

The model we adopted for the amplifier in the last section – that of a static, zero-order system with constant gain – is over-simplified, since we know there is a significant dynamic behaviour in any feedback amplifier; we therefore propose a more realistic dynamic model with s-domain representation given by:[30]

$$G(s) = \frac{A(s)}{1+H(s)A(s)}\,.$$

According to the argument above, we must avoid the condition $1 + H(s)A(s) = 0$.

17.2. Zeros and poles

Writing the transfer function as a general fraction

$$G(s) = \frac{P(s)}{Q(s)}\,,$$

we can make the following associations:

- Values of s such that $P(s) = 0$ are *zeros*, i.e. $G(s) \to 0$ as $P(s) \to 0$;
- Values of s such that $Q(s) = 0$ are *poles*, i.e. $G(s) \to \infty$ as $Q(s) \to 0$.

In real (physical) systems the transfer function $G(s)$ is *always* a ratio of two polynomials with real coefficients. This results in the poles and zeros being either real *or* complex-conjugate pairs (recall the previous definition $s = \sigma + j\omega$). We can check for stability by seeing where the roots of $Q(s)$ lie on the complex plane. Generally,

- Roots in the $\sigma < 0$ side of the plane are stable;
- Roots in the $\sigma > 0$ side of the plane are unstable.

17.3. Roots of the transfer function

Sometimes roots will be complex and, as stated above, in real physical systems these always appear as complex conjugate pairs. Generally, as we have seen, we can expand the transfer function as

$$G(s) = \sum_{k=1}^{n} \frac{A_k}{s - s_k}\,.$$

This is a simplification in order to illustrate the point, though in practice many real-world transfer functions are actually expandable in this form. The complex-conjugate roots will be of the form

$$s_k = \sigma_k \pm j\omega_k\,,$$

so we obtain a set of solutions of the form

$$g_k(t) = A_k\, e^{s_k t} + A_k^*\, e^{s_k^* t}\,.$$

Since $A_k = a + jb$ and $s_k = \sigma_k + j\omega_k$ we can simplify to

$$\begin{aligned} g_k(t) &= 2\sqrt{a^2+b^2}\, e^{\sigma_k t}\cos(\omega_k t + \beta_k) \\ \beta_k &= \tan^{-1}\frac{b}{a}\,. \end{aligned}$$

These results are illustrated in Figure 17.1. In the top panel, a single, real negative root exists and the system is stable. In the second panel, the system is unstable as the output grows without limit. In the third panel complex-conjugate roots exist to the left of the $\sigma = 0$ axis; the result is damped oscillation and the system is stable. With complex roots on the RHS of the axis we have growing

[30] Following the third rule for block diagram algebra given at the end of Section 15.

oscillation and the system is unstable. The last two panels show some special cases. With purely imaginary roots we have sustained oscillation, which is nonetheless stable. If the root is at $s = 0$ then the response is a constant, which is stable.

Note that *all* roots must be on the $\sigma < 0$ side of the plane for stability. *Any one root* on the right of the plane will give instability.

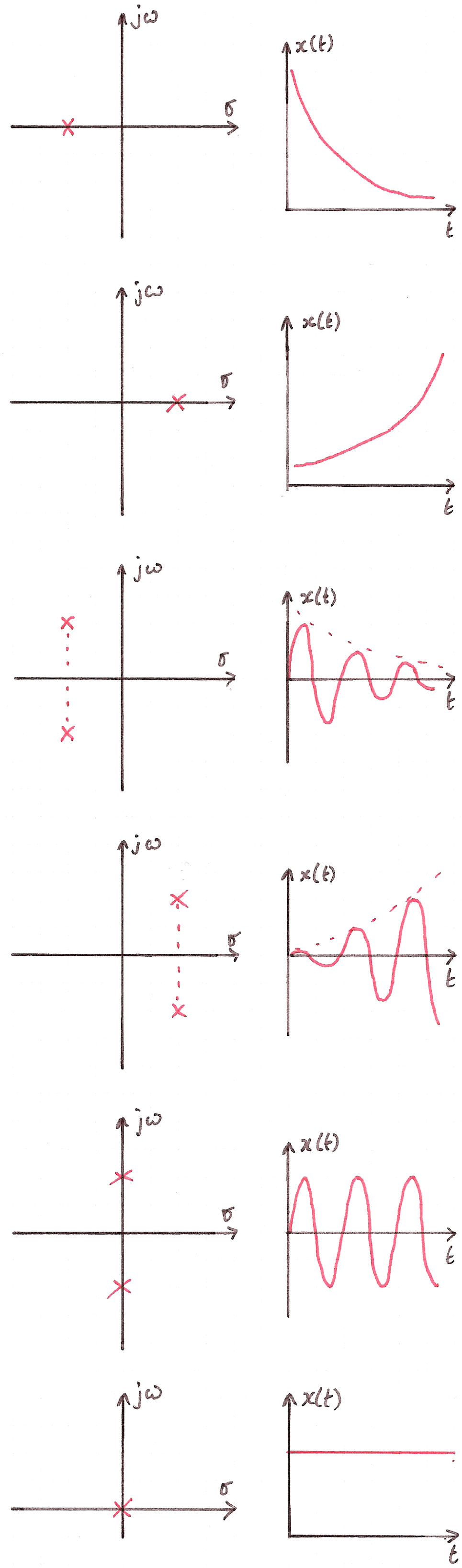

Figure 17.1: Graphical illustration of roots in the complex plane.

18. Systems with initial conditions

We will now use the Laplace transform to analyse a simple electrical circuit. One of the major uses of the Laplace transform in electrical engineering is for initial value problems where we throw a switch, or turn on a circuit. What we then want to know is the behaviour of the circuit for all times after we throw the switch.

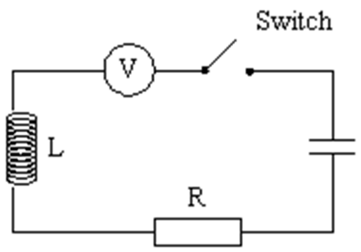

Figure 18.1: Switched RLC circuit.

Figure 18.1 shows the simple RLC series circuit with a voltage source V and a switch. If the voltage source were a sine wave generator, and the switch had been closed for a long time, then we could use our usual expressions for the complex impedance (essentially a Fourier approach) to find the current and voltage for various components in the circuit. However, here the applied voltage is a constant V and is applied once at time $t = 0$. AC circuit analysis is not appropriate for this sort of transient input problem where we want to know "what happens immediately after we throw the switch?" We can use the Laplace transform to answer this.

Consider the following problem: for a DC voltage source $V = 300$ V, inductance $L = 2$ H, capacitance $C = 0.02$ F and resistance $R = 16\ \Omega$, we want to find an expression for the charge on the capacitor $q(t)$ and the current $i(t)$ in the circuit for all times $t > 0$ after the switch is closed. Assume that the circuit is initially relaxed.

We begin by determining the differential equation which governs the circuit. We can write the voltage across each of the components as

- Voltage across the inductor $= L\frac{\mathrm{d}i}{\mathrm{d}t}$;
- Voltage across the resistor $= iR$;
- Voltage across the capacitor $= \frac{q}{C}$.

Using Kirchhoff's voltage law to sum the voltages around the series circuit we can write (for time $t > 0$, i.e. *after* the switch is closed):

$$L\frac{\mathrm{d}i}{\mathrm{d}t} + iR + \frac{q}{C} = V\,. \tag{18.1}$$

Since $i = \mathrm{d}q/\mathrm{d}t$ (we know that the instantaneous current is the same everywhere in the circuit, due to conservation of charge, or Kirchhoff's Current Law) and also $V = 0$ for all time $t < 0$, we can write this as

$$Vu(t) = L\frac{\mathrm{d}^2q}{\mathrm{d}t^2} + R\frac{\mathrm{d}q}{\mathrm{d}t} + \frac{q}{C} \tag{18.2}$$

where $u(t)$ is the unit step function. The differential equation is second-order in time, therefore we may expect that the system may be capable of oscillation as well as tending to some steady-state condition at some time t much after we close the switch. To solve the DE we take the Laplace transform:

$$\begin{aligned}\frac{V}{s} &= L\left(s^2Q(s) - sq(0^+) - \frac{\mathrm{d}q(0^+)}{\mathrm{d}t}\right)\\ &\quad + R(sQ(s) - q(0^+))\\ &\quad + \frac{Q(s)}{C}\,.\end{aligned} \tag{18.3}$$

Here we see how the initial conditions of the system are automatically taken care of by the Laplace transform. Since the circuit is initially relaxed, we start with $q = 0$ and $\dot{q} = \ddot{q} = 0$, so we can further simplify the algebraic expression

$$\frac{300}{s} = 2s^2Q(s) + 16sQ(s) + 50Q(s)\,,$$

$$Q(s) = \frac{150}{s(s^2+8s+25)}\,.$$

Again, using partial fractions, here we must write

$$\frac{150}{s(s^2+8s+25)} = \frac{A}{s} + \frac{Bs+C}{s^2+8s+25}\,.$$

Multiplying by s and setting $s = 0$ yields $A = 6$. Multiplying by $s(s^2+8s+25)$, taking $\mathrm{d}/\mathrm{d}s$ and setting $s = 0$ yields $B = -6$. Again, taking $\mathrm{d}/\mathrm{d}s$ and setting $s = 0$ yields $C = -48$:

$$Q(s) = \frac{150}{s(s^2+8s+25)} = \frac{6}{s} - \frac{6s+48}{s^2+8s+25}\,.$$

This requires further simplification before we can use the standard transforms:

$$\begin{aligned}Q(s) &= \frac{6}{s} - \frac{6(s+4)+24}{(s+4)^2+9}\\ &= \frac{6}{s} - \frac{6(s+4)}{(s+4)^2+9} - \frac{24}{(s+4)^2+9}\,.\end{aligned}$$

We can now use entries 2, 11 and 12 in Figure 15.2 to get:

$$q(t) = 6 - 6e^{-4t}\cos(3t) - 8e^{-4t}\sin(3t)\,.$$

Now we have $q(t)$ we can differentiate to find the current:

$$i(t) = \frac{\mathrm{d}q}{\mathrm{d}t} = 50e^{-4t}\sin(3t)\,.$$

This is illustrated in Figure 18.2. Note that this is the kind of response we expect from a damped harmonic oscillator. The result is physically meaningful: at the instant at which the switch is closed, the rate of change of voltage across the capacitor is very large, so a large current will flow, with the rate of change of current limited by the inductor. Electrical energy is stored in the electric field across the capacitor and in the magnetic field of the inductor. The two exchange energy in an oscillatory fashion like a mechanical oscillator exchanging kinetic and potential energy. The $\sin 3t$ term reflects this. As the system evolves, electrical energy is converted to heat energy in the resistor, so the amplitude of the response decays (the exponential term) – and, in fact, it overshoots into negative territory a little. For large t the applied voltage (the forcing function V) is a constant, and the voltage across the capacitor tends to a constant – hence the impedance of the capacitor tends to ∞ and the current tends to zero.

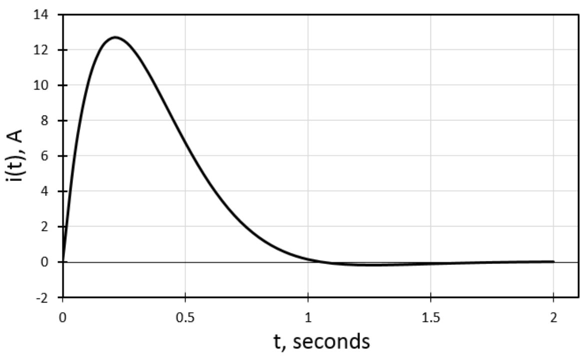

Figure 18.2: Time response of the switched RLC circuit.

In this example we started from the understanding that the system was "initially relaxed". This is usually the starting point for most systems. When we switch-on the Elvis prototyping board, we assume that the voltages and currents are zero, and this is usually right; however, consider in this case that someone has snuck into the lab overnight and charged the capacitor up to 100 V. The charge on the capacitor will stay put until the switch is closed, and we can imagine that the $i(t)$ response will be very different! The defining differential equation for the circuit is still valid (see equations (18.1) and (18.2) and we can see that the transformation into the s-domain (equation (18.3)) automatically includes an initial value for q.